



Name: Cohort/Batch: Date: Score:

ZOOKEEPER PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Distributed Systems

TOTAL: 10 QUESTIONS

Q1 What is ZooKeeper and what problem in Distributed Systems does it solve?

Q6 How does ZooKeeper handle reliability and high availability?

Q2 What are the core components or concepts in ZooKeeper?

Q7 What observability does ZooKeeper provide out of the box?

Q3 How does ZooKeeper compare to its main alternatives?

Q8 What are common performance bottlenecks in ZooKeeper?

Q4 How do you get started with ZooKeeper for a new project?

Q9 What security best practices apply when running ZooKeeper?

Q5 What configuration options most affect ZooKeeper's behavior in production?

Q10 What mistakes do teams commonly make when adopting ZooKeeper?



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 ZooKeeper is a tool or platform that

Mitigation

ZooKeeper is a tool or platform that automates or simplifies a specific concern in the Distributed Systems domain, reducing manual effort and operational complexity for engineering teams.

A6 ZooKeeper provides HA through leader election, replication, or stateless horizontal scaling.

Availability

ZooKeeper provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 ZooKeeper has key building blocks that work

Primitives

ZooKeeper has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 ZooKeeper exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces.

Observability

ZooKeeper exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 ZooKeeper trades off simplicity against feature richness

Hardening

ZooKeeper trades off simplicity against feature richness differently than its alternatives. Choose ZooKeeper when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches.

Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install ZooKeeper via its package manager or binary release.

Gotchas

Install ZooKeeper via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run ZooKeeper with the principle of least privilege

Assessment

Run ZooKeeper with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels.

Configuration

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.

Documentation

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.